

# Nicholas M. Johnson

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## EDUCATION

**Cornell University, College of Engineering**, Ithaca, NY

Bachelor of Engineering - BE, Computer Science

May 2026

## TECHNICAL SKILLS

**Skills:** Python • Typescript • Java • C++ • OCaml • Solidity • React • Tailwind • FastAPI • NodeJS • PostgreSQL • Docker • PyTorch • TensorFlow • Scikit-learn • OpenCV • RAG • Agentic AI • Figma • Clay

## RELEVANT EXPERIENCE

**Cornell Rocketry**, *Recovery and Payload Team*, Ithaca, NY

September 2021 – May 2024

- Engineered autopilot flight software integrating GPS, altimeter, compass, and accelerometer data to guide payload recovery to a predefined landing zone
- Designed and implemented a solar panel deployment and positioning system to recharge flight batteries
- Worked alongside a large team to win the Spaceport America Cup

**Versa Finance**, *Founder*, San Francisco, CA

February 2020 – January 2021

- Prototyped personal finance app leveraging bank account data, machine learning, and behavioral psychology to curb spending and increase savings
- Led a team of 12 in the creation of a business plan, marketing plan, psychological validation, and machine learning bank statement analysis

**Infrared Air**, *Founder*, Boulder, CO

August 2019 – May 2020

- Engineered a drone platform integrating RGB and thermal imaging to identify and diagnose defective solar cells
- Conducted inspections of 100+ residential and commercial solar arrays

## LEADERSHIP EXPERIENCE

**Flight Discovery Program**, *Founder*, Lyons, CO

August 2019 – May 2020

- Designed hands on school curriculum to educate and inspire students in the field of flight
- Taught classes on aerodynamics, drone piloting, and rocketry, culminating in student-designed rocket launches

**Light My Fire Camp**, *Founder*, Lyons, CO

April 2019 – June 2019

- Founded and organized a camp to match middle school students with high school students with similar interests to work on projects of their choosing
- Fundraised to make camp participation and projects fully free for all accepted students

## ADDITIONAL EXPERIENCE

**LifeTote**, *Co-Founder*, Boulder, CO

December 2021 – September 2024

- Prototyped and engineered a convertible tote bag that rapidly transforms into a bulletproof vest
- Negotiated with suppliers and local manufacturers to reduce cost and shorten lead times
- Launched company website and Kickstarter campaign, developed business plan, and secured patent protection
- Scaled revenue to \$50k and Improved product quality while cutting manufacturing costs by 60%

**East Penn Manufacturing**, *Material Handler*, Boulder, CO

December 2024 – June 2025

- Executed warehouse operations, including order packing and fulfillment and interacted with customers across Colorado

## PROJECT EXPERIENCE

**Solid-Guard**, *Explorations in ML*, Cornell University, Ithaca, NY

September 2025 – January 2026

- Engineered a reproducible smart contract security dataset pipeline that generates paired malicious and safe Solidity contracts, runs Slither static analysis, and analyzes for common attack vulnerabilities

**Clairvoyant Crime Detection**, *Independent*, Ithaca, NY

January 2026 – May. 2026

- Built an AI-powered video surveillance search engine using React, TypeScript, and FastAPI, enabling natural language queries across multi-camera feeds with sub-300ms retrieval via PGVector ANN search
- Engineered a cross-camera entity re-identification system with confidence scoring, real-time clip playback

## INTERESTS

Rocketry • Fixed-Wing FPV • Drone Racing • Guitar • Boxing